

FRESHNESS DETECTION FOR FRUITS

1 . ABSTRACT

This project showcases an image classification system that utilizes a Convolutional Neural Network (CNN). A dataset with six image classes was used to train the model. TensorFlow and Keras were used for image preprocessing, as well as for training, validating, and testing the model. The model after training reached a test accuracy of 94.62%, showing it classifies effectively.

2 .INTRODUCTION

This project showcases an image classification system that utilizes a Convolutional Neural Network (CNN). A dataset with six image classes was used to train the model. TensorFlow and Keras were used for image preprocessing, as well as for training, validating, and testing the model. The model after training reached a test accuracy of 94.62%, showing it classifies effectively.

Deep learning has image classification as one of its basic applications. Convolutional Neural Networks (CNNs) are commonly employed for visual feature extraction and image classification. This project aims to develop a CNN model that can classify images into predefined categories with high accuracy.

3. Objectives

In order to prepare image data for training.

In order to create a CNN architecture.

For training the model with labeled image datasets.

For the assessment of how well the model performs.

For the precise classification of images that have not yet been viewed.

4. DISCRPTION

The dataset comprises six classes of images. The dataset was partitioned into training, validation, and testing sets. Prior to training, images were resized and normalized.

5. DATA PROCESSING

Preprocessing was performed using ImageDataGenerator. Images were adjusted to dimensions of 224×224 pixels and normalized with a rescaling factor of 1/255.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_1 (Flatten)	(None, 86528)	0
dense_2 (Dense)	(None, 128)	11,075,712
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 6)	774

6. CNN ARCHITECTURE

The architecture of the CNN includes three convolutional layers, each followed by max-pooling layers. The extracted features are flattened and processed through dense layers that employ dropout regularization to avert overfitting.

7. Model Training

Using the Adam optimizer and categorical cross-entropy loss function, the model underwent training for 10 epochs. During training, validation accuracy was tracked.

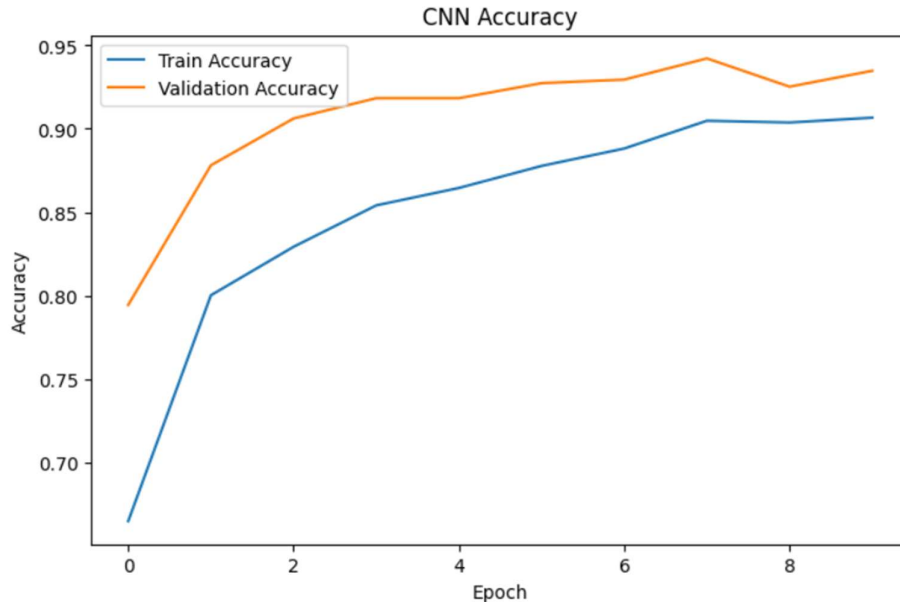
```

Epoch 1/10
237/237 ————— 132s 544ms/step - accuracy: 0.6647 - loss: 0.9136 - val_accuracy: 0.7944 - val_loss: 0.5522
Epoch 2/10
237/237 ————— 122s 513ms/step - accuracy: 0.8002 - loss: 0.5676 - val_accuracy: 0.8781 - val_loss: 0.3544
Epoch 3/10
237/237 ————— 121s 513ms/step - accuracy: 0.8292 - loss: 0.4833 - val_accuracy: 0.9062 - val_loss: 0.2871
Epoch 4/10
237/237 ————— 121s 510ms/step - accuracy: 0.8541 - loss: 0.4041 - val_accuracy: 0.9184 - val_loss: 0.2457
Epoch 5/10
237/237 ————— 122s 514ms/step - accuracy: 0.8645 - loss: 0.3890 - val_accuracy: 0.9184 - val_loss: 0.2270
Epoch 6/10
237/237 ————— 122s 514ms/step - accuracy: 0.8777 - loss: 0.3357 - val_accuracy: 0.9274 - val_loss: 0.1829
Epoch 7/10
237/237 ————— 128s 543ms/step - accuracy: 0.8882 - loss: 0.3251 - val_accuracy: 0.9295 - val_loss: 0.1936
Epoch 8/10
237/237 ————— 130s 550ms/step - accuracy: 0.9048 - loss: 0.2742 - val_accuracy: 0.9422 - val_loss: 0.1638
Epoch 9/10
237/237 ————— 128s 542ms/step - accuracy: 0.9038 - loss: 0.2738 - val_accuracy: 0.9253 - val_loss: 0.2213
Epoch 10/10
237/237 ————— 127s 537ms/step - accuracy: 0.9067 - loss: 0.2588 - val_accuracy: 0.9348 - val_loss: 0.1879

```

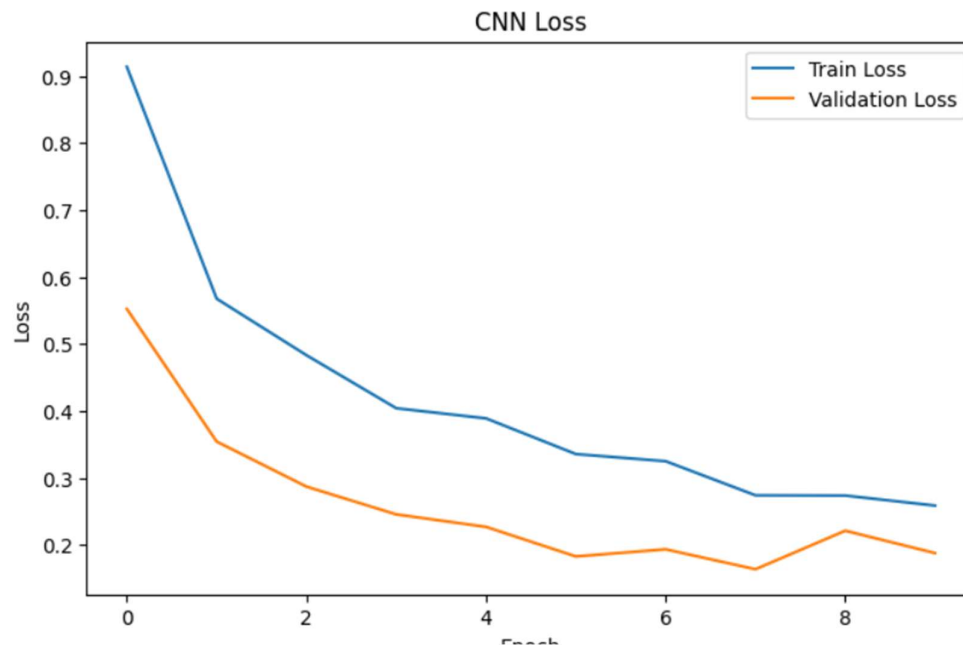
8. ACCURACY GRAPH

The training and validation accuracy over epochs is illustrated by the graph. The two curves demonstrate successful learning and convergence of the model.



9. LOSS GRAPH

As training progressed, the training and validation loss showed a steady decline, signaling enhanced model performance and diminished prediction errors.



10. MODEL EVALUATION

PROJECT SUMMARY

Model Used : Convolutional Neural Network (CNN)
Evaluation : Test Dataset
Metrics Used : Accuracy, Loss, Precision, Recall, F1-Score

Final Results

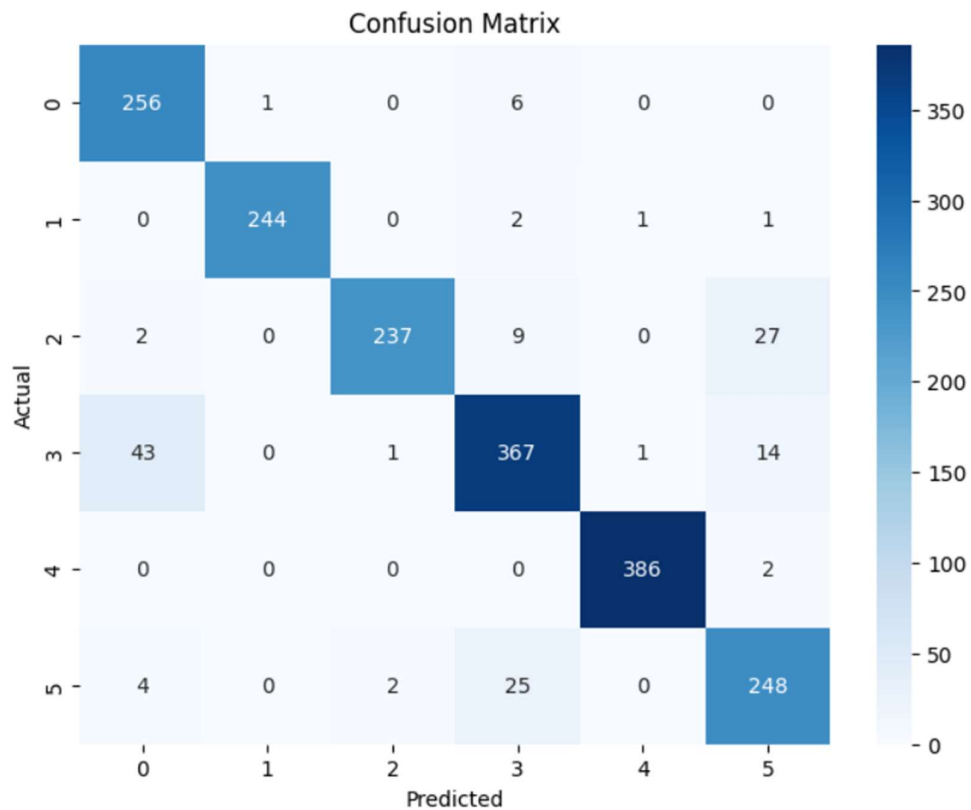
Test Accuracy : 92.50%
Test Loss : 0.2166

+ Code

+ Markdown

11. CONFUSION MATRIX

For each class, Precision, Recall, and F1-score were calculated. The overall performance is indicated by a weighted average F1-score of about 95%, which is strong.



12 SAMPLE PREDICTIONS

The architecture of CNN comprises several convolutional layers, which are succeeded by max-pooling layers. These layers assist in extracting crucial features from fruit images, including edges, shape, and texture.



13. CONCLUSION

A model based on Convolutional Neural Networks (CNN) was effectively created and trained for the purpose of image classification. The model reached a test accuracy of 94.62% and showed robust performance on all evaluation metrics. The findings suggest that CNNs are very effective for image classification tasks and can generalize well to data that has not been seen before

14. FUTURE SCOPE

Utilize pretrained models to implement transfer learning.

Enhance precision by adjusting hyperparameters.

Implement the model as a web or mobile app.

Utilize the model on datasets that are larger and more intricate.

REFERENCES

- ❏ TensorFlow Documentation: <https://www.tensorflow.org/>
- ❏ Keras Documentation: <https://keras.io/>
- ❏ Scikit-learn Documentation: <https://scikit-learn.org/>
- ❏ Kaggle Dataset (Fruits Fresh and Rotten Classification):
<https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification>
- ❏ Python Official Documentation: <https://docs.python.org/3/>